

The Rural Mobilization Gap in U.S. Place-Based Tax Credit:

Intermediary Selection versus Market Structure in the New Markets Tax Credit*

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Abstract

The U.S. New Markets Tax Credit (NMTC), enacted in 2000, subsidizes private investors with a 39% federal tax credit conditional on routing capital through certified Community Development Entities (CDEs) into projects in low-income census tracts. Over FY2001–FY2022 the program deployed \$66.6 billion of public credit into 8,024 projects worth \$120.9 billion, implying an aggregate private-mobilization ratio of \$0.82 per federal dollar. Non-metropolitan tracts received 19.6% of dollars—effectively the 20% statutory minimum—but project-level leverage in non-metro tracts is systematically lower (median $1.07\times$ versus $1.19\times$ metro). This paper decomposes the rural mobilization penalty using project-level fixed-effects regressions on the CDFI Fund’s public release. The naive estimate is a precise -0.262 rural penalty in mean leverage. Year, project-type, and state fixed effects shrink it to -0.172 . Adding CDE fixed effects moves the coefficient to -0.047 ; the mean regression is imprecise (cluster-robust 95% CI $[-0.245, +0.151]$), but the median-regression analog is a precise -0.001 (SE 0.008), and within intermediary both the extensive margin (any private mobilization) and the intensive

*Contact: ianthelfrich@gmail.com. Working-paper draft of August 13, 2026; comments welcome. Data, code, figures, and an interactive viewer are public at <https://github.com/ihelfrich/us-nmtc-viewer>. All errors are mine.

margin are null. An order-invariant Gelbach decomposition attributes -0.185 of the raw gap to CDE identity (CDE-cluster bootstrap 95% CI $[-0.347, -0.023]$; 86% of the explained movement), against -0.041 for project type. The aggregate rural leverage gap is therefore largely a between-CDE composition fact, not a within-CDE deployment penalty. A cross-CDE bunching test on the 20% non-metropolitan statutory share finds no excess mass at the cumulative-deployment level (excess mass $\hat{B} = -0.0006$; cluster-bootstrap 95% CI $[-0.014, +0.013]$), consistent with the mandate binding at the allocation-award level rather than the deployment level. The findings reframe the rural mobilization debate: the operative policy levers concern intermediary selection and capacity, not market-structure redesign.

JEL: H25, H81, R51, G28.

Keywords: New Markets Tax Credit; place-based policy; blended finance; mobilization ratio; Community Development Entities; rural finance; fixed-effects decomposition.

1 Introduction

The blended-finance literature has long debated which structures most effectively mobilize private capital alongside public concessional finance (Convergence, 2024; Attridge & Engen, 2019; Lankes, 2021). A persistent empirical limitation runs through that debate: the “mobilization ratio”—private dollars moved per public dollar committed—is rarely observed directly at the project level. Most evidence relies on survey-reported figures or on coarse program-level aggregates.

The U.S. New Markets Tax Credit is a setting where this central blended-finance quantity is *directly observed*. For every project, the CDFI Fund’s public data release records both the federally subsidized investment (the Qualified Low-Income Community Investment, QLICI) and the total project cost. Their ratio is exactly the leverage that the literature otherwise infers indirectly. The NMTC is therefore an unusually clean within-country laboratory for measurement-driven blended-finance research: a single instrument, a single country, twenty-two years of deployment, and roughly 8,000 projects.

The question this paper asks is simple to state. The NMTC mobilizes less private capital per public dollar in non-metropolitan tracts than in metropolitan ones. Is that gap a *market-structure* phenomenon—rural projects are fundamentally less leverageable—or an *intermediary-selection* phenomenon—the Community Development Entities that deploy in rural tracts are systematically weaker mobilizers everywhere they operate? The distinction carries direct policy content. A market-structure story argues for redesigning the credit; a selection story argues for reallocating it.

The answer, in this data, is composition. The raw rural penalty in project-level leverage is -0.262 and precisely estimated. Layering in origination-year, project-type, and state fixed effects moves it to -0.172 . Adding CDE fixed effects—comparing rural and urban deals *made by the same intermediary*, a comparison to which 163 of 343 CDEs contribute, covering 94% of rural projects—collapses the point estimate to -0.047 . This paper is deliberate about what that null does and does not establish: the mean regression is imprecise (cluster-robust 95% CI $[-0.245, +0.151]$), and on its own can reject only within-CDE penalties larger than 0.21. The precision lives elsewhere, and the paper reports it where it lives: the median within-CDE penalty is -0.001 (SE 0.008), rural deals within a CDE are no less likely to mobilize any private capital at all, and mobilize no less when they do. What differs across space is *which* intermediaries show up.

This paper makes three contributions to a literature that has examined NMTC outcomes extensively but mobilization sparingly (Freedman, 2012; Harger & Ross, 2016;

Theodos et al., 2022). First, to my knowledge it is the first econometric paper to treat project-level leverage as the central NMTC outcome variable, importing the blended-finance mobilization framework into a literature otherwise focused on neighborhood employment, demographic change, and price effects. Second, it implements a layered fixed-effects decomposition with CDE fixed effects, separating the within-CDE rural penalty from the between-CDE selection component—a decomposition that descriptive work by Theodos et al. (2021) motivates but does not estimate. Third, it applies a Chetty et al. (2011)/Kleven & Waseem (2013) style bunching diagnostic to the cross-CDE distribution of non-metro deployment shares around the program’s 20% statutory minimum—a test that, despite the mandate’s prominence in NMTC policy discussion, has not previously been run. The test finds no excess mass at the 20% line in cumulative deployment (excess mass -2.2% of the counterfactual; a CDE-level bootstrap places the 95% confidence interval squarely around zero), which is consistent with compliance binding at the allocation-award level that the public release does not report.

The rest of the paper proceeds as follows. Section 2 describes the program’s mechanics and the institutional heterogeneity of the CDE intermediary layer. Section 3 describes the public data release and the construction of the leverage outcome. Section 4 lays out the layered fixed-effects design, its interpretation, and its limits. Section 5 presents the decomposition, heterogeneity, robustness, and the bunching diagnostic. Section 6 draws out the policy reading, and Section 7 concludes. A causal companion design—a regression-discontinuity analysis at the low-income-community eligibility threshold—requires a tract-level ACS merge and is deferred to follow-on work.

2 Institutional background

2.1 Program mechanics

The NMTC, enacted in the Community Renewal Tax Relief Act of 2000, awards investors a 39% federal tax credit, claimed over seven years, on Qualified Equity Investments (QEIs) made into certified Community Development Entities. CDEs in turn deploy that capital as Qualified Low-Income Community Investments into Qualified Active Low-Income Community Businesses (QALICBs) located in eligible census tracts. The public release distinguishes four QALICB types: real-estate projects (RE), non-real-estate operating businesses (NRE), special-purpose entities (SPE), and CDE-to-CDE loans. The credit’s mechanics make the CDE the program’s pivotal actor: it holds the

allocation, originates the deals, and structures the capital stack around the subsidized tranche.

2.2 Eligibility and the 20% non-metro minimum

Eligible low-income community (LIC) tracts are defined by a poverty rate of at least 20% or median family income at or below 80% of the area benchmark, with a targeted-populations override. Relevant here, the program’s authorizing statute (P.L. 106–554, §121) directs that non-metropolitan counties receive a proportional share of investments, implemented in practice as a 20% non-metro target. Aggregate deployment sits almost exactly at that line: non-metro tracts received 19.6% of QLICI dollars over the sample period.

2.3 The intermediary layer

Roughly 600 CDEs were active at least once over 2001–2022; 310 executed five or more transactions. Concentration is high—the top twenty CDEs account for more than half of dollars—and specialization is extreme: cross-CDE non-metro deployment shares range from 0% to more than 80%. This heterogeneity is the engine of the paper’s decomposition. If rural deals systematically flow through intermediaries who mobilize less *everywhere*, an aggregate rural penalty can arise without any within-intermediary rural disadvantage. The public release identifies CDEs by name; institutional-form classification (bank subsidiary, nonprofit CDFI, for-profit specialist) is a planned extension.

3 Data

The analysis uses the CDFI Fund’s NMTC Public Data Release covering FY2003–FY2022 award years (released June 2024), which reports 19,907 QLICI transactions across 8,024 projects with origination years 2001–2022. Transactions are aggregated to the project level; the project is the unit at which total cost—and therefore leverage—is defined. Cleaning steps, field definitions, and file hashes are documented in the public repository’s data dictionary and provenance files, and the full pipeline from the raw workbook to every table in this paper runs from a single script sequence.

Table 1: NMTC deployment by metro status, FY2001–FY2022.

	Metro	Non-metro
Projects	6,463	1,561
QLICI deployed (\$M)	53,566	13,045
Total project cost (\$M)	97,151	23,751
Mean leverage	1.99	1.73
Median leverage	1.19	1.07

The outcome is project-level leverage,

$$\text{Leverage}_i = \frac{\text{ProjectCost}_i}{\text{QLICI}_i},$$

the blended-finance mobilization ratio plus one. Leverage is bounded below at 1 by construction; the mass at exactly 1 has the substantive interpretation of zero private mobilization. The baseline outcome is winsorized to $[1, 20]$ to discipline a long right tail of large capital-stack deals; Section 5.4 shows the headline result is insensitive to the cap, including its removal.

Table 1 reports descriptives. Non-metro projects are 19.5% of the sample—again, the statutory line—and lag metro projects in both mean (1.73 versus 1.99) and median (1.07 versus 1.19) leverage. Figure 1 shows the shape of the gap: non-metro mass piles up near the 1.0 floor while the metro distribution carries a fatter right shoulder.

4 Empirical strategy

4.1 Layered fixed-effects decomposition

Let R_i indicate a non-metro project. The paper estimates the sequence

$$L_i = \alpha + \beta R_i + \delta_{t(i)} + \eta_{q(i)} + \mu_{s(i)} + \gamma_{c(i)} + \varepsilon_i,$$

adding origination-year (δ), QALICB-type (η), state (μ), and CDE (γ) fixed effects in turn. Each layer removes a candidate composition story. The CDE step is the workhorse: with $\gamma_{c(i)}$ included, $\hat{\beta}$ is identified from rural–urban comparisons *within* intermediary, so it measures the within-CDE rural penalty—the market-structure component. The shrinkage of $\hat{\beta}$ upon adding CDE fixed effects measures the between-CDE selection

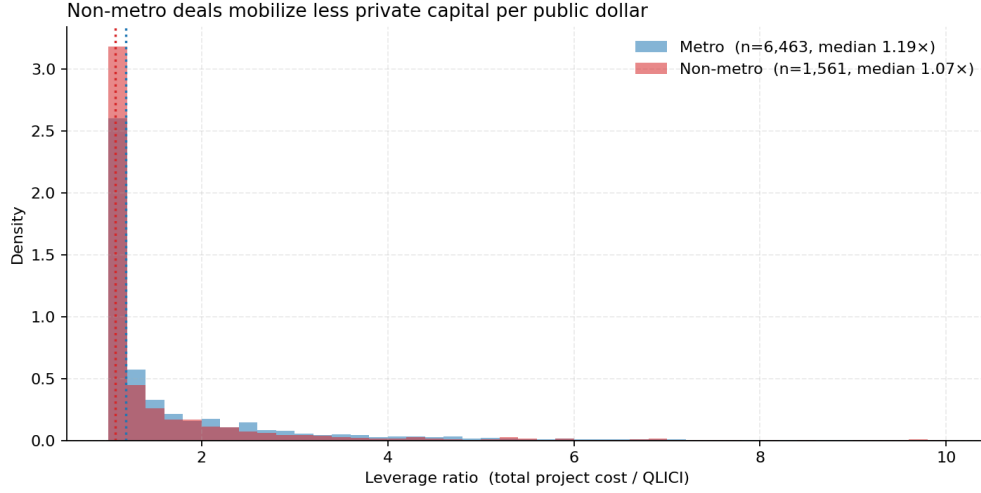


Figure 1: Project-level leverage distributions by metro status. Non-metro mass concentrates at the 1.0 $\times$ floor (zero private mobilization); the metro distribution has the fatter right shoulder.

component. Standard errors in CDE-fixed-effects specifications are clustered at the CDE level, the level of the fixed effect and the economic decision-maker.

4.2 Interpretation and limits

The decomposition is descriptive in the sense of selection-on-observables: within-CDE allocation of deals across rural and urban tracts is not random, and no LATE interpretation of $\hat{\beta}$ is claimed. What the design does establish is an accounting fact with economic content: whether the aggregate rural penalty survives comparisons inside the intermediary, where deal-structuring capacity, underwriting practice, and investor relationships are held fixed. A causal companion design—fuzzy RDD at the LIC eligibility threshold, run separately for metro and non-metro samples—requires tract-level ACS covariates and is deferred to the follow-on paper.

One further caution belongs here, because it constrains the language the paper is entitled to use. The between-CDE component is a composition fact; its “selection” reading is an interpretation. Rural market conditions could in principle cause low-leverage intermediaries to specialize in rural deployment, in which case part of the between-CDE component is market structure operating through intermediary business models. The paper’s secure claim is the within-CDE null; the decomposition locates the gap, and the discussion section is explicit about which readings of its location the data can and

cannot separate.

4.3 The 20% mandate as a notch

For each CDE j , define its cumulative non-metro deployment share $s_j = n_{j,\text{nonmetro}}/n_{j,\text{total}}$ over 2001–2022. If the statutory 20% minimum binds at the deployment margin, optimizing CDEs should pile up at $s_j = 0.20$, producing excess mass in the Chetty et al. (2011)/Kleven (2016) sense. The test compares empirical mass in a ± 2.5 percentage-point window around 20% against a degree-3 polynomial counterfactual fitted outside the window, over the 310 CDEs with at least five transactions, with a CDE-level bootstrap for inference.

5 Results

5.1 The decomposition

Table 3 is the paper’s central exhibit. The raw rural gap (column 1) is -0.262 with a heteroskedasticity-robust standard error of 0.060 : non-metro projects mobilize about a quarter of a dollar less private capital per public dollar, and the difference is precise. Origination-year effects barely move it (column 2, -0.249): the gap is not a vintage story. QALICB-type effects (column 3) shrink it to -0.186 —rural deals skew toward operating-business types with structurally lower leverage, and that composition explains roughly a quarter of the raw gap. State effects (column 4) move it modestly further to -0.172 .

Column 5 adds CDE fixed effects. The rural coefficient collapses to -0.047 with a CDE-clustered standard error of 0.101 : an order of magnitude smaller than the raw gap in point-estimate terms, though the interval ($[-0.245, +0.151]$) is wide, and Section 5.2 quantifies exactly how much the mean regression can and cannot rule out. The median specification (column 6) is the precise one: the within-CDE rural penalty at the median is -0.001 (SE 0.008).

Because sequential addition of fixed-effect blocks is order-dependent, Table 2 reports the Gelbach (2016) decomposition with the full model as reference, which is invariant to listing order. Of the -0.216 movement from the raw to the full specification, CDE identity contributes -0.185 (86% of the movement; CDE-cluster bootstrap 95% CI $[-0.347, -0.023]$), QALICB type -0.041 , and origination year a small offsetting $+0.010$.

Table 2: Gelbach (order-invariant) decomposition of the raw rural gap, full-model reference. Contributions sum to $\hat{\beta}_{M0} - \hat{\beta}_{\text{full}}$ by identity (verified to 10^{-14} in the replication code).

	Contribution	Share of movement
Origination year	+0.010	-4.8%
QALICB type	-0.041	19.2%
CDE identity	-0.184	85.6%
Total ($\hat{\beta}_{M0} - \hat{\beta}_{\text{full}}$)	-0.216	100%
CDE component bootstrap 95% CI: [-0.347, -0.023] (499 CDE-cluster reps)		

Table 3: The rural leverage gap under layered fixed effects. Outcome: project leverage, winsorized [1, 20]. Columns (1)–(4) report HC1 standard errors; columns (5)–(6) cluster at the CDE level. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	(1) OLS	(2) OLS	(3) OLS	(4) OLS	(5) OLS	(6) Median
Non-metro ($\hat{\beta}$)	-0.262*** (0.060)	-0.249*** (0.061)	-0.186*** (0.061)	-0.172** (0.067)	-0.047 (0.101)	-0.001 (0.008)
Year FE	—	✓	✓	✓	✓	✓
QALICB-type FE	—	—	✓	✓	✓	✓
State FE	—	—	—	✓	—	—
CDE FE	—	—	—	—	✓	✓
R^2	0.002	0.015	0.022	0.039	0.131	—
N	8,024	8,024	8,024	8,024	8,024	8,024

The share version of this statement—82% of the raw gap—carries a wide bootstrap interval ([25%, 255%]) because it is a ratio of estimates, and the paper therefore rests on the level claim: the CDE composition component is large, negative, and statistically distinct from zero, while the within-CDE component is statistically indistinguishable from zero.

5.2 Precision, power, and margins

The mean and median estimates differ in precision by an order of magnitude, and the paper states plainly what each can support. For the mean specification, a one-sided 5% test can reject within-CDE rural penalties larger than 0.213—nearly the size of the raw gap—so the mean regression alone is weak evidence for a zero; the outcome’s long right tail, which inflates mean-regression variance, is the proximate reason. The median specification can reject penalties larger than 0.013. The paper’s null is therefore carried

by the median estimate and by the margins, with the mean reported for comparability. (The median standard error is the quantile-regression asymptotic one; a CDE-clustered bootstrap for the median with roughly six hundred fixed effects is computationally disproportionate for this draft and is noted as an outstanding item.)

The margins address a second way a mean could mislead: 27.4% of projects sit exactly at the leverage floor of one, meaning zero private mobilization, so a within-CDE penalty could in principle hide in the probability of mobilizing at all. It does not. Within CDE, rural deals are 1.8 percentage points less likely to clear the floor (SE 1.8pp, $p = 0.34$), and among deals that mobilize, the rural coefficient is -0.039 ($p = 0.77$). Both margins are stable when the floor is moved to 1.05. Identification for all within-CDE results comes from the 163 of 343 CDEs that deploy on both sides of the rural line; these switchers originate 94% of all rural projects, so the comparison is broad, not a boutique subsample.

5.3 Heterogeneity by project type

Interacting the rural indicator with QALICB type (with CDE and year fixed effects retained) yields uniformly imprecise interactions: the rural main effect is 0.091 (SE 0.621), and no type-specific rural penalty is distinguishable from zero. The within-CDE null is not concentrated in, nor masked by, any single project type. The point estimates weakly echo the descriptive pattern—most negative for real estate, where private debt-stacking is most natural—but the data cannot reject homogeneity.

5.4 Robustness

Table 4 probes the workhorse specification. Unwinsorizing the outcome makes the naive gap *larger* (-0.338 , $p = 0.01$) and the within-CDE estimate remains null (-0.215 , $p = 0.48$): the winsorization is not producing the headline result. A log outcome puts the within-CDE penalty at -3.2% ($p = 0.18$). Tightening or loosening the winsorization cap ($[1, 10]$, $[1, 50]$), splitting the sample at 2010, and restricting to single-CDE projects all leave the within-CDE rural coefficient statistically indistinguishable from zero. Restricting to the fifty largest CDEs, where the fixed effects are estimated most precisely, gives -0.100 (SE 0.148); two-way clustering by CDE and census tract moves the headline standard error from 0.101 to 0.105. The null is not an artifact of outcome trimming, era, stacked-allocation deals, subsample, or clustering choice.

Table 4: Robustness of the within-CDE rural coefficient. All rows except the first include year, QALICB-type, and CDE fixed effects with CDE-clustered standard errors. *** $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.

	$\hat{\beta}$	(SE)	p	N
Unwinsorized outcome, no FE	-0.338**	(0.136)	0.01	8,024
Unwinsorized outcome	-0.215	(0.304)	0.48	8,024
Log outcome	-0.032	(0.024)	0.18	8,024
Winsorized at [1, 10]	-0.066	(0.073)	0.37	8,024
Winsorized at [1, 50]	-0.047	(0.138)	0.73	8,024
Origination 2001–2009	-0.044	(0.147)	0.77	2,488
Origination 2010–2022	-0.039	(0.107)	0.71	5,536
Single-CDE projects only	-0.088	(0.139)	0.53	6,366
Top-50 CDEs only	-0.100	(0.148)	0.50	4,229
Two-way cluster (CDE $\times$ tract)	-0.047	(0.105)	0.66	8,024
Bunching excess mass $\hat{B}$	-0.0006	95% CI [-0.0136, 0.0135], 999 CDE-bootstrap reps		

5.5 No bunching at the 20% line

Figure 2 plots the cross-CDE distribution of cumulative non-metro shares against the polynomial counterfactual. Excess mass in the window around the 20% statutory line is $\hat{B} = -0.0006$ —that is, *negative* and -2.2% of the counterfactual mass—with a CDE-level bootstrap 95% confidence interval of $[-0.014, +0.013]$; 47% of bootstrap draws are positive. There is no evidence that CDEs manage their cumulative deployment to the mandate line.

Two readings are consistent with this null. First, the mandate may bind at the allocation-award level—commitments CDEs make in their allocation agreements—which the public release does not report; cumulative realized deployment would then sit wherever award-level compliance leaves it. Second, the cross-CDE distribution is close to bimodal, with urban specialists near zero and rural specialists far above 20%, leaving the mandate line in a low-density valley between modes. Both readings imply the same caution for policy discussion: the 20% line, prominent as it is in NMTC debate, is not where deployment decisions pile up.

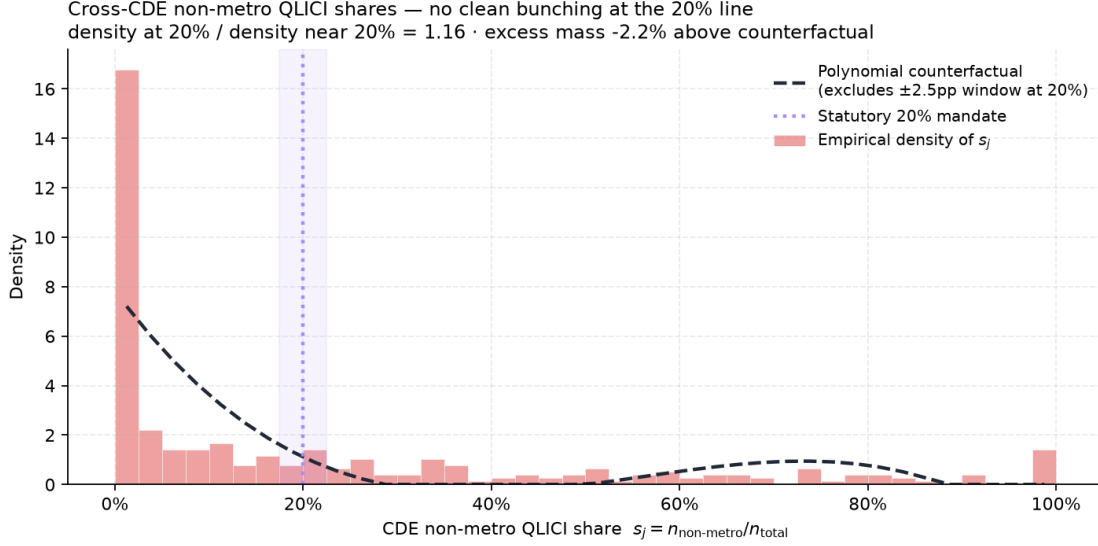


Figure 2: Cross-CDE distribution of cumulative non-metro deployment shares s_j , CDEs with at least five transactions ($n = 310$). The dashed line is a degree-3 polynomial counterfactual fitted excluding the shaded ± 2.5 pp window at the 20% statutory line.

6 Discussion

6.1 What the results do and do not say

The results say that the within-CDE rural penalty is precisely zero at the median, null at both the extensive and intensive margins, and small but imprecisely estimated in the mean; and that the aggregate rural penalty—which is real—is located in between-CDE composition. They do not say that rural credit markets are no different from urban ones; they say the NMTC’s deployment data do not detect such a difference once the intermediary is held fixed. They also do not fully adjudicate *why* composition looks as it does: reading the between-CDE component as intermediary selection is the natural interpretation, but rural market conditions could in principle drive low-leverage intermediaries to specialize in rural deployment, which would put part of the between-CDE component back on the market-structure side of the ledger. Distinguishing those readings requires variation in CDE–market assignment that the public release does not contain. Nor, finally, do the results say that any particular CDE could not mobilize more in rural tracts; the estimates are averages over the fixed-effects distribution.

6.2 Policy reading

Three implications follow. First, the marginal rural deal does not appear constrained by market structure under the existing program design: when a given intermediary goes rural, its mobilization does not measurably suffer. Second, the aggregate rural gap is therefore amenable to *intermediary-allocation* levers—shifting allocation toward CDEs with demonstrated high-leverage capacity, independent of their current rural orientation—without redesigning the credit itself. Third, the 20% mandate as currently observed operates on a margin (cumulative deployment share) where behavior shows no bunching; if the policy goal is rural mobilization rather than rural deal counts, the mandate is pointed at the wrong margin.

6.3 Research program

This paper is the first phase of a sequence: the NMTC alone here; a multi-program U.S. comparison (NMTC, LIHTC, Opportunity Zones) next; and an international extension to development-bank project finance where the same mobilization framework applies. The LIC-eligibility regression discontinuity, which requires the ACS tract merge, is the immediate companion piece.

7 Conclusion

Directly observed project-level leverage in the NMTC shows a real aggregate rural mobilization penalty that is not visible inside the intermediary: the order-invariant decomposition places -0.185 of the raw -0.262 gap on CDE identity, the within-CDE penalty is precisely zero at the median and null at both margins, and the mean estimate, while imprecise, is an order of magnitude below the raw gap. The cross-CDE distribution of non-metro shares shows no bunching at the 20% statutory line. Subject to the interpretive caution that intermediary composition may itself respond to rural market conditions, the rural mobilization debate, at least for this program, is better posed in intermediary terms than in market-structure terms, and the operative levers are allocation levers.

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A Reproducibility

Every number in this draft regenerates from the public repository: `scripts/describe_nmtc.py` builds the analysis files from the raw CDFI workbook (hashes in `PROVENANCE.md`); `scripts/run_regressions.py` produces Table 3 and the bunching statistics; `scripts/run_robustness.py` produces Table 4; `scripts/make_paper_tables.py` writes the table fragments included here. The repository also serves a browser-based viewer with the full project map and specification explorer.